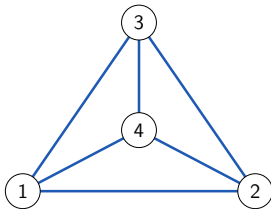


Planarity

Section 1.5

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



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Today's lecture

1. Definitions and Examples

2. Euler's Formula

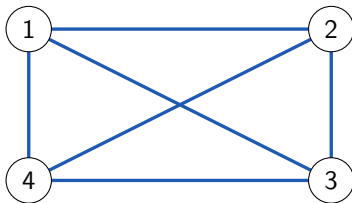
3. Regular Polyhedra

4. Kuratowski's Theorem

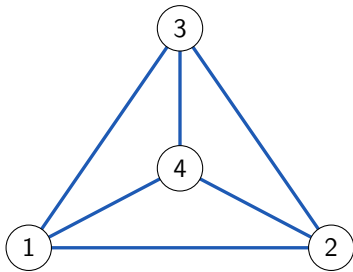
Can We Draw This Graph Without Crossings?

Question

The drawing below has crossing edges. Does the graph have to be drawn this way?



The Same Graph, Drawn Differently



The graph is still K_4 : every pair of vertices is adjacent.

We have changed the drawing, not the graph.

Definition: Planar Graph and Plane Graph

Planar graph

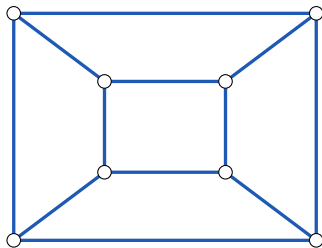
A graph is **planar** if it has a drawing in the plane in which edges meet only at common endpoints.

Plane graph

A particular crossing-free drawing is a **plane embedding** of the graph.

A crossing in one drawing does not establish nonplanarity.

Example: A Plane Drawing of the Cube



Question

Count its vertices and edges. How many regions does the drawing divide the plane into?

Definition: Regions and Boundaries

Region (face)

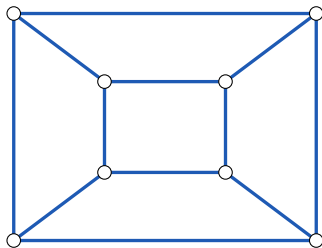
A connected component of the plane left after removing the embedded graph. The unbounded component is also a region.

Boundary length

The number of edge occurrences in a walk around the boundary, counted with multiplicity.

A bridge can occur twice on the boundary of one region.

Counting the Regions of the Cube



There are 8 vertices, 12 edges, and 6 regions, including the outside.

$$8 - 12 + 6 = 2.$$

Why Boundary Edges Must Be Counted with Multiplicity

Example

A tree with m edges has just one region. Each edge has both sides on that region, so its boundary length is $2m$.

Each edge has two sides. They may belong to two different regions or to the same region.

$$\sum_R b(R) = 2m.$$

Theorem: Euler's Formula

Euler's Formula

For a connected plane graph with n vertices, m edges, and f regions,

$$n - m + f = 2.$$

The number of regions is therefore determined by the abstract graph once a plane embedding exists.

Proof of Euler's Formula: The Tree Case

Recall

A tree with n vertices has $n - 1$ edges.

Proof

A plane tree has no cycle enclosing a region, so it has only the unbounded region. Thus $f = 1$, and

$$n - m + f = n - (n - 1) + 1 = 2.$$

This includes the single-vertex tree.

Proof Continued: Removing a Cycle Edge

Proof

If the connected plane graph is not a tree, choose an edge e on a cycle. It is not a bridge, so deleting it preserves connectedness.

Its two sides border distinct regions. Deleting e merges those regions: m and f both decrease by one, while n is unchanged.

Hence $n - m + f$ is unchanged. Repeat until a tree remains; its value is 2. □

Euler's Formula: A Worked Example

Question

A connected plane graph has 12 vertices and 18 edges. How many regions does it have?

$$f = 2 - n + m = 2 - 12 + 18 = 8.$$

If another crossing-free drawing of the same graph looks very different, it still has eight regions.

Disconnected Plane Graphs

Extension

If a plane graph has c connected components, then

$$n - m + f = 1 + c.$$

Proof

Join the components by $c - 1$ new edges without crossings. Each new edge is a bridge and creates no new region. Apply Euler's formula to the resulting connected graph:

$$n - (m + c - 1) + f = 2.$$

Rearrange.



Counting Edge Sides

Boundary identity

For every plane graph,

$$\sum_R b(R) = 2m.$$

In a connected simple plane graph with $n \geq 3$, every region has boundary length at least 3.

$$3f \leq 2m.$$

This combines a local observation about faces with Euler's global formula.

Theorem: An Edge Bound for Planar Graphs

Planar edge bound

A simple planar graph with $n \geq 3$ vertices has at most $3n - 6$ edges.

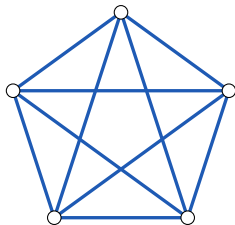
Proof

First assume connectedness and choose a plane embedding. Euler gives $f = 2 - n + m$, while $3f \leq 2m$. Thus

$$3(2 - n + m) \leq 2m, \quad m \leq 3n - 6.$$

If the graph is disconnected, add crossing-free edges to connect it and apply the connected case. □

The Complete Graph on Five Vertices



Question

Could a different drawing remove all crossings?

K_5 has $n = 5$ and $m = 10$, whereas $3n - 6 = 9$.

Therefore K_5 is not planar.

A Stronger Bound Without Triangles

Triangle-free planar edge bound

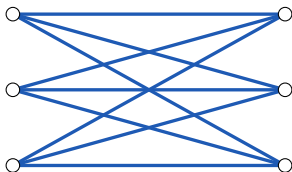
If G is simple, planar, triangle-free, and $n \geq 3$, then

$$m \leq 2n - 4.$$

Proof

In a connected plane embedding, every facial boundary has length at least four. Thus $4f \leq 2m$. Substitute $f = 2 - n + m$ and rearrange. Disconnected graphs can be joined by bridges without creating triangles. □

The Three Utilities Problem



Question

Can each of three houses be connected to each of three utilities without any crossings?

The required graph is $K_{3,3}$.

Why the Three Utilities Cannot Be Connected

Proof

$K_{3,3}$ is bipartite, so it has no triangle. It has $n = 6$ vertices and $m = 9$ edges. If it were planar, the triangle-free bound would give

$$9 \leq 2(6) - 4 = 8,$$

a contradiction. □

The obstruction is independent of how the houses and utilities are placed.

The Edge Bound Is Necessary, Not Sufficient

Question

Does $m \leq 3n - 6$ imply that a graph is planar?

No. $K_{3,3}$ satisfies $9 \leq 12$, but is not planar.

We need structural obstructions as well as numerical bounds.

Theorem: A Vertex of Small Degree

Degree bound

Every nonempty simple planar graph has a vertex of degree at most 5.

Proof

The cases $n \leq 2$ are immediate. For $n \geq 3$,

$$\frac{1}{n} \sum_v \deg(v) = \frac{2m}{n} \leq 6 - \frac{12}{n} < 6.$$

If every degree were at least six, the average would be at least six. □

What Happens When Equality Holds?

Triangulations

For a connected simple plane graph with $n \geq 3$, equality $m = 3n - 6$ holds exactly when every facial boundary has length three.

Proof

The only inequality in the edge-bound proof was $3f \leq 2m$. Equality means every term in $\sum_R b(R)$ is exactly 3. □

A Bound Using the Girth

Girth

The length of a shortest cycle.

Girth bound

For a connected simple plane graph containing a cycle and having girth g ,

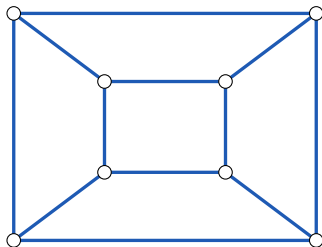
$$m \leq \frac{g}{g-2}(n-2).$$

Proof

Every facial boundary has length at least g , so $gf \leq 2m$. Combine this with $f = 2 - n + m$ and solve for m . □

Graphs on Polyhedra

A convex polyhedron determines a plane graph: choose a face as the outside and project its edges onto the plane.



Vertices, edges, and faces correspond to those of the polyhedron.

Euler's Formula for Polyhedra

Polyhedral Euler formula

A convex polyhedron with V vertices, E edges, and F faces satisfies

$$V - E + F = 2.$$

Example

For a cube: $V = 8$, $E = 12$, $F = 6$. For a tetrahedron: $V = 4$, $E = 6$, $F = 4$.

Definition: Regular Polyhedron

Regular polyhedron

A convex polyhedron whose faces are congruent regular polygons, with the same number of faces meeting at every vertex.

Suppose each face has p edges and each vertex has degree q .

$$pF = 2E, \quad qV = 2E.$$

Both p and q are at least 3.

Classifying the Possible Regular Polyhedra

Substitute $V = 2E/q$ and $F = 2E/p$ into Euler's formula:

$$\frac{2E}{q} - E + \frac{2E}{p} = 2.$$

Consequently,

$$\frac{1}{p} + \frac{1}{q} > \frac{1}{2}.$$

Question

Which integer pairs $p, q \geq 3$ satisfy this inequality?

Only Five Pairs Are Possible

If $p, q \geq 4$, then $1/p + 1/q \leq 1/2$. Thus one equals 3.

If $p = 3$, the inequality gives $q < 6$, so $q = 3, 4, 5$.

Interchanging p and q gives the remaining possibilities:

$$(3, 3), (3, 4), (4, 3), (3, 5), (5, 3).$$

The Five Platonic Solids

Solid	p	q	V	E	F
Tetrahedron	3	3	4	6	4
Cube	4	3	8	12	6
Octahedron	3	4	6	12	8
Dodecahedron	5	3	20	30	12
Icosahedron	3	5	12	30	20

The inequality proves there are no other possibilities; these five solids realize the five pairs.

A Small Face Must Exist

Small-face observation

Every convex polyhedron has a face with at most five sides.

Proof

Each vertex has degree at least three, so $3V \leq 2E$. If every face had at least six sides, then $6F \leq 2E$. Hence

$$V - E + F \leq \frac{2E}{3} - E + \frac{E}{3} = 0,$$

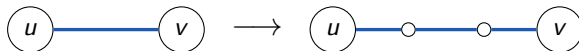
contradicting Euler's formula.



Definition: Subdivision

Subdivision

Replace an edge uv by a path from u to v , inserting new vertices of degree two. Repeating this operation gives a subdivision of the graph.



Subdivision preserves planarity.

Homeomorphic Graphs

Homeomorphic

Two graphs are homeomorphic if they have isomorphic subdivisions.

In drawings, a degree-two vertex can be inserted along an edge or suppressed along a path without creating a crossing.

This lets us recognize a nonplanar configuration even when its edges have been stretched into longer paths.

Theorem: Kuratowski's Theorem

Kuratowski

A finite graph is planar exactly when it contains no subgraph that is a subdivision of K_5 or $K_{3,3}$.

The subdivision need not be induced: extra edges can be ignored.

We use the theorem as a structural criterion; its full sufficiency proof is beyond this lecture.

Why the Forbidden Subdivisions Obstruct Planarity

Proof

A subgraph of a planar graph is planar. If a subdivision of K_5 or $K_{3,3}$ were planar, suppressing its inserted degree-two vertices would give a plane drawing of K_5 or $K_{3,3}$.

We have proved that neither graph is planar. Therefore no planar graph can contain either subdivision. □

A Subdivision Can Pass the Numerical Tests

Example

Subdivide every edge of $K_{3,3}$ once. The resulting graph has 15 vertices and 18 edges.

It satisfies both $18 \leq 3(15) - 6$ and $18 \leq 2(15) - 4$.

Nevertheless, it is nonplanar: it is itself a subdivision of $K_{3,3}$.

How to Recognize a Subdivision

- ① Identify the branch vertices: five for K_5 , or two groups of three for $K_{3,3}$.
- ② Replace each required edge by a path between its endpoints.
- ③ Check that these paths have pairwise disjoint interiors.
- ④ Delete irrelevant edges and vertices.

Paths may meet at required branch vertices, but not at additional internal vertices.

Practice: Prove or Disprove

Question

- (a) Every graph with at most $3n - 6$ edges is planar.
- (b) Every subgraph of a planar graph is planar.
- (c) Adding a vertex of degree one preserves planarity.
- (d) Every planar graph has maximum degree at most five.

Practice: Answers

- (a) False: $K_{3,3}$.
- (b) True: delete the unwanted parts of a plane drawing.
- (c) True: place the new vertex close to its neighbor.
- (d) False: the star $K_{1,r}$ is planar for every r . The theorem bounds the *minimum* degree, not the maximum.

Summary: Planarity

- A planar graph admits a crossing-free drawing.
- Euler's formula links vertices, edges, and regions.
- Face counting gives edge and degree bounds.
- Euler's formula restricts regular polyhedra to five types.
- Kuratowski's theorem gives the two subdivision obstructions.

Colorings